

The postulates of Quantum Mechanics

Postulate 1

1. A physical system is associated to a Hilbert space $\mathcal{H}$. The state of the system at time t is represented by a state vector $|\psi(t)\rangle$ of that Hilbert space.

$|\psi(t)\rangle$ contains all information about the system.

superposition: $|\phi\rangle = c_1|\psi\rangle + c_2|\chi\rangle$

basis of a two-dimensional Hilbert space: $\{|\uparrow\rangle, |\downarrow\rangle\}$

$$|\psi\rangle = a|\uparrow\rangle + b|\downarrow\rangle$$

$$|\phi\rangle = a|\uparrow\rangle - b|\downarrow\rangle$$

different bases for the same Hilbert space: $\{|+\rangle, |-\rangle\}$

$$\{|\odot\rangle, |\oslash\rangle\}$$

$$|\uparrow\rangle = \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle)$$

$$|\downarrow\rangle = \frac{1}{\sqrt{2}}(|+\rangle - |-\rangle)$$

Postulate 1

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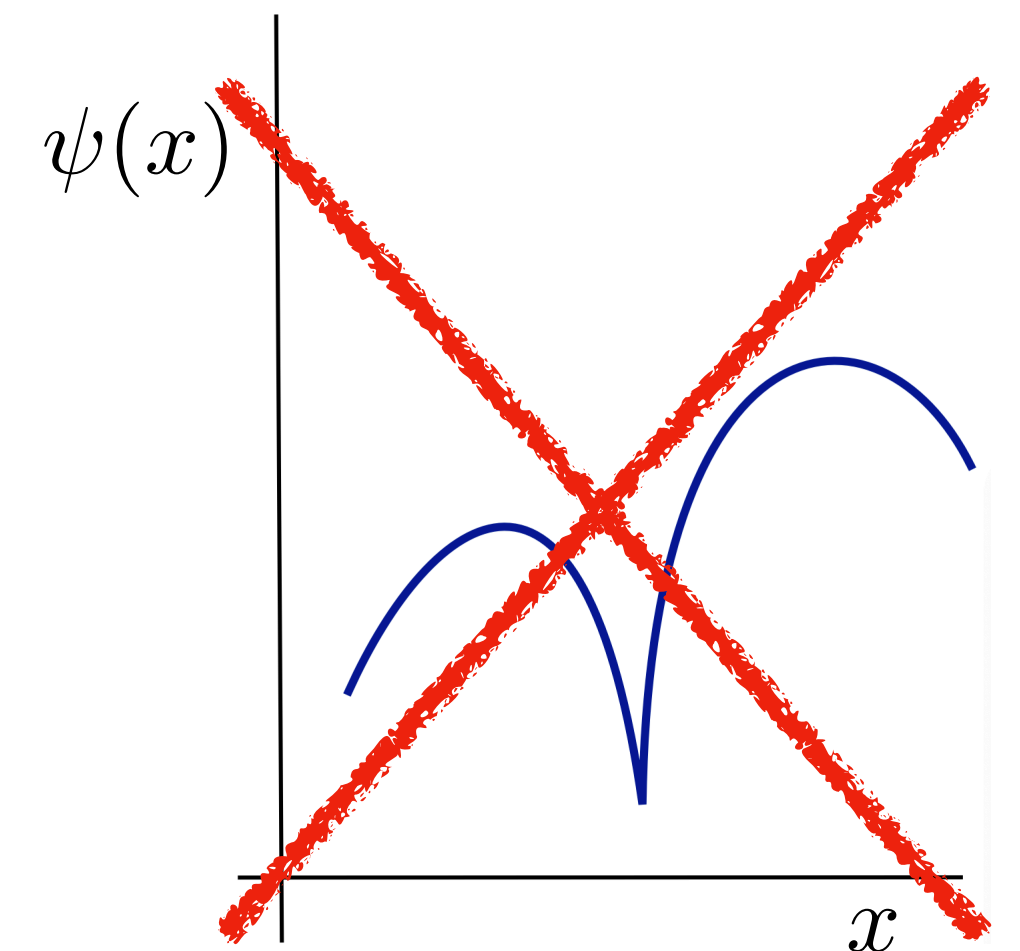
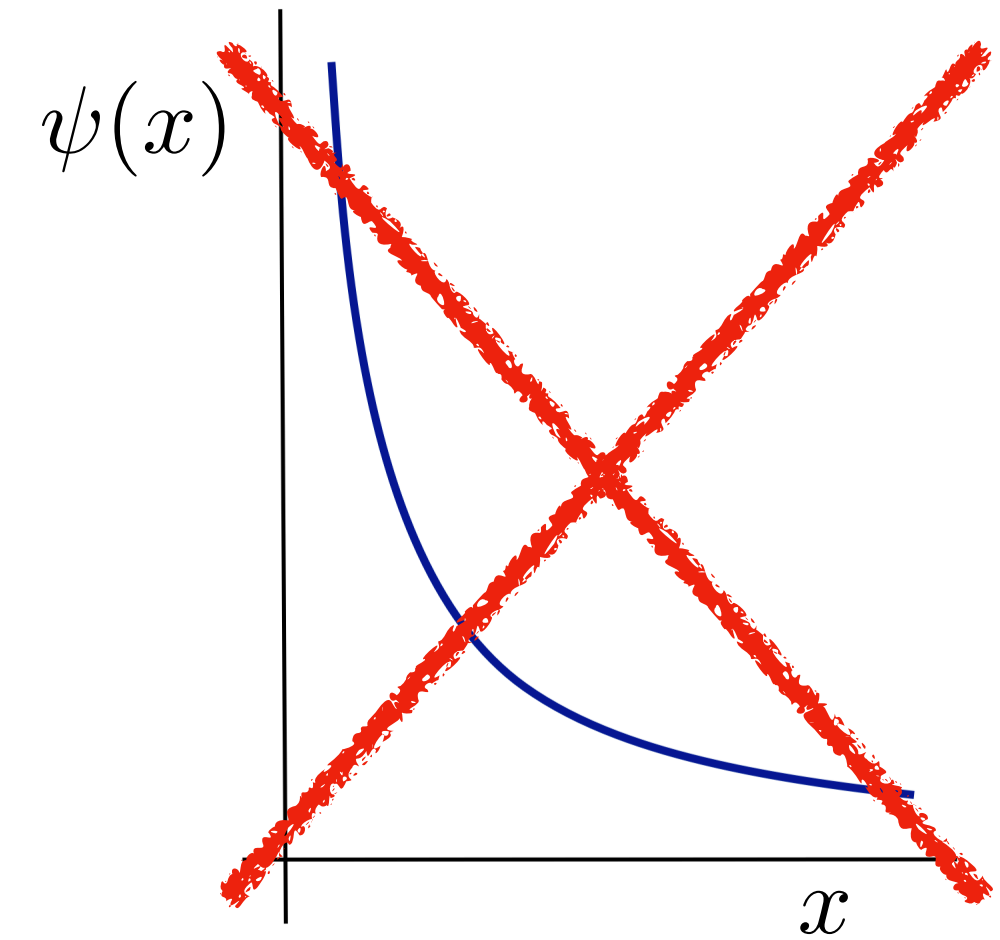
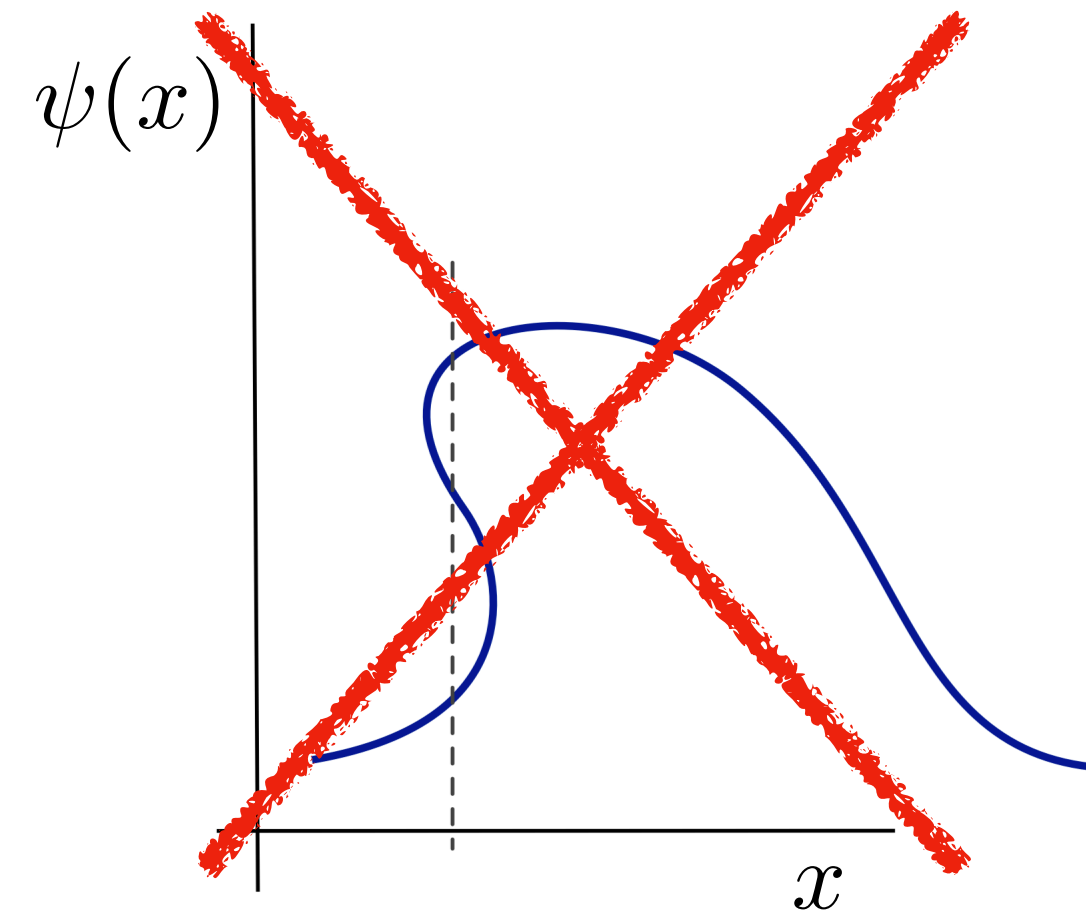
$|\psi(t)\rangle$ contains all information about the system.

Physically acceptable wave functions:

- finite
- single valued
- continuous
- same conditions for the first derivative

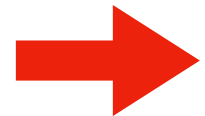
Born interpretation: $|\Psi(r, t)|^2 d^3r = P(r, t)$

$$\int |\Psi(r, t)|^2 d^3r = 1$$



Postulate 2

2. Each physically measurable quantity A has associated a Hermitian linear operator $\hat{A}$, whose eigenvectors form a complete basis of a Hilbert space.

A Hermitian operator has real eigenvalues:  physically measurable quantities correspond to real numbers

position:

$$\vec{r} \rightarrow \hat{r}$$

linear momentum:

$$\vec{p} = m\vec{v} \rightarrow \hat{p} = -i\hbar\vec{\nabla}$$

energy:

$$H = \frac{1}{2m}\vec{p}^2 + V(r, t) \rightarrow \hat{H} = -\frac{\hbar^2}{2m}\vec{\nabla}^2 + \hat{V}(r, t)$$

angular momentum:

$$\vec{L} = \vec{r} \times \vec{p} \rightarrow \hat{L} = -i\hbar \hat{r} \times \vec{\nabla}$$

Postulate 3

3. The only possible outcome of a measurement of an observable A associated to the operator $\hat{A}$ is one of the eigenvalues a_n of the operator. The probability of measuring a_n for a system described by state vector $|\psi\rangle$ is given by

$$p_n = \langle\psi|\hat{P}_n|\psi\rangle = \langle\psi|(|\phi_n\rangle\langle\phi_n|)|\psi\rangle$$

➔ Example: $|\psi\rangle = |c_1\phi_1 + c_2\phi_2 + c_3\phi_3\rangle$

$$p_1 = \langle c_1\phi_1 + c_2\phi_2 + c_3\phi_3|\phi_1\rangle\langle\phi_1|c_1\phi_1 + c_2\phi_2 + c_3\phi_3\rangle = |c_1|^2$$

➔ If the state vector $|\psi\rangle$ corresponds to an eigenvector of $\hat{A}$ the only possible outcome of a measurement is a_n

$$|\psi\rangle = |\phi_n\rangle \Rightarrow p_n = 1$$

➔ The sum of probabilities for obtaining any eigenvalue is 1

$$\sum_n p_n = \sum_n \langle\psi|(|\phi_n\rangle\langle\phi_n|)|\psi\rangle = \langle\psi|\psi\rangle = 1$$

Postulate 3

3. The only possible outcome of a measurement of an observable A associated to the operator $\hat{A}$ is one of the eigenvalues a_n of the operator.

Spectrum of eigenvalues: discrete or continue

$$\begin{array}{llll} \hat{A}|\psi_n\rangle = a_n|\psi_n\rangle & \text{if } a_n \in \mathbb{Z} & \rightarrow & \text{discrete spectrum of eigenvalues} \\ & a_n \in \mathbb{R} & \rightarrow & \text{continuous spectrum of eigenvalues} \end{array}$$

Example 1: $\hat{A} = \hat{P}_x = -i\hbar \frac{\partial}{\partial x}$

$$-i\hbar \frac{\partial}{\partial x} e^{ikx} = (\hbar k) e^{ikx}$$

$\hbar k$ can take any real value, hence **continuous** spectrum of eigenvalues

Example 2: $\hat{A} = \hat{L}_x = -i\hbar \frac{\partial}{\partial \phi}$ with $0 \leq \phi < 2\pi$ and $\lim_{\phi \rightarrow 2\pi} \psi(\phi) = \psi(0) \Rightarrow e^{im2\pi} = e^{im0} = 1$

$$-i\hbar \frac{\partial}{\partial \phi} e^{im\phi} = (\hbar m) e^{im\phi}$$

$\hbar m$ only when $m = 0, \pm 1, \pm 2, \dots$
 m must be integer, hence **discrete** spectrum of eigenvalues

Postulate 4

4. If a physical system is described by a state $|\psi\rangle$ and one measures A obtaining a_n , the system immediately after the measurement changes to ϕ_n

➔ In other words, $|\psi\rangle$ is projected onto the eigenvector corresponding to the measured eigenvalue

$$|\psi'\rangle = (|\phi_n\rangle\langle\phi_n|)|\psi\rangle = a_n|\phi_n\rangle$$

➔ Measuring A for a system described by $|\psi\rangle$ changes the state vector to $|\psi'\rangle$

$$\hat{A}|\psi\rangle = |\psi'\rangle$$

which can be rewritten as $\hat{A}|\psi\rangle = a_n|\phi_n\rangle$, describing the action a measurement.

Postulate 4

4. If a physical system is described by a state $|\psi\rangle$ and one measures A obtaining a_n , the system immediately after the measurement changes to ϕ_n

➡ The mean value of a large number of measurements is known as the expectation value

$$\langle A \rangle = \sum_n a_n p_n = \sum_n a_n \langle \psi | (|\phi_n\rangle \langle \phi_n|) | \psi \rangle = \sum_n a_n |c_n|^2$$

More commonly written as

$$\langle A \rangle = \langle \psi | \hat{A} | \psi \rangle \quad \longleftarrow \quad \text{substitute: } |\psi\rangle = \sum_i c_i |\phi_i\rangle$$

$$= \langle \sum_i c_i \phi_i | \hat{A} | \sum_j c_j \phi_j \rangle$$

$$= \sum_i c_i^* \sum_j c_j \langle \phi_i | \hat{A} | \phi_j \rangle$$

$$= \sum_i \sum_j c_i^* c_j a_j \langle \phi_i | \phi_j \rangle = \sum_j |c_j|^2 a_j$$

Postulate 5

5. The time evolution of a state vector $|\psi(t)\rangle$ is governed by the time-dependent

Schrödinger equation
$$i\hbar \frac{\partial |\psi(t)\rangle}{\partial t} = \hat{H}(t) |\psi(t)\rangle$$

➔ The norm of $|\psi(t)\rangle$ is preserved during the time evolution

$$i\hbar \frac{d}{dt} \langle \psi(t) | \psi(t) \rangle = \left[i\hbar \frac{d\langle \psi(t) |}{dt} \right] |\psi(t)\rangle + \langle \psi(t) | \left[i\hbar \frac{d|\psi(t)\rangle}{dt} \right] = -\langle \psi(t) | \hat{H}(t) | \psi(t) \rangle + \langle \psi(t) | \hat{H}(t) | \psi(t) \rangle = 0$$

➔ Expectation values change (in general) with time

$$\begin{aligned} \frac{d}{dt} \langle \psi(t) | \hat{A} | \psi(t) \rangle &= \left[\frac{d\langle \psi(t) |}{dt} \right] \hat{A} | \psi(t) \rangle + \langle \psi(t) | \hat{A} \left[\frac{d|\psi(t)\rangle}{dt} \right] + \langle \psi(t) | \frac{\partial \hat{A}}{\partial t} | \psi(t) \rangle \\ &= -\frac{i}{\hbar} \langle \psi(t) | [\hat{A}, \hat{H}(t)] | \psi(t) \rangle + \langle \psi(t) | \frac{\partial \hat{A}}{\partial t} | \psi(t) \rangle \end{aligned}$$

Postulate 5

➔ Time evolution operator: $|\psi(t)\rangle = \hat{U}(t, t_0)|\psi(t_0)\rangle$

substitution in the Schrödinger equation $i\hbar \frac{\partial |\psi(t)\rangle}{\partial t} = \hat{H}(t)|\psi(t)\rangle$ gives $i\hbar \frac{\partial}{\partial t} \{\hat{U}(t, t_0)|\psi(t_0)\rangle\} = \hat{H}(t)\{\hat{U}(t, t_0)|\psi(t_0)\rangle\}$

since $|\psi(t_0)\rangle$ does not depend on t: $\frac{\partial \hat{U}(t, t_0)}{\partial t} |\psi(t_0)\rangle = -\frac{i}{\hbar} \hat{H}(t) \hat{U}(t, t_0) |\psi(t_0)\rangle$

these two operators
are equal

for a time-independent Hamiltonian $\frac{\partial \hat{U}(t, t_0)}{\partial t} = -\frac{i}{\hbar} \hat{H} \hat{U}(t, t_0)$ can be integrated to give

➔ $\hat{U}(t, t_0) = e^{-i(t-t_0)\hat{H}/\hbar}$ and $|\psi(t)\rangle = e^{-i(t-t_0)\hat{H}/\hbar} |\psi(t_0)\rangle$

Postulate 5

$$i\hbar \frac{\partial |\psi(t)\rangle}{\partial t} = \hat{H}(t) |\psi(t)\rangle$$

Time independent potentials: Stationary states

Schrödinger equation in position representation: $i\hbar \frac{\partial \psi(\vec{r}, t)}{\partial t} = -\frac{1}{2} \nabla^2 \psi(\vec{r}, t) + \hat{V}(\vec{r}) \psi(\vec{r}, t)$

➡ separable solutions: $\psi(\vec{r}, t) = f(t) \phi(\vec{r})$


$$i\hbar \frac{\partial f(t)}{\partial t} \phi(\vec{r}) = -f(t) \frac{1}{2} \nabla^2 \phi(\vec{r}) + f(t) \hat{V}(\vec{r}) \phi(\vec{r})$$

divide by $f(t) \phi(\vec{r})$:

$$i\hbar \frac{1}{f(t)} \frac{\partial f(t)}{\partial t} = \frac{1}{\phi(\vec{r})} \left[-\frac{1}{2} \nabla^2 \phi(\vec{r}) + \hat{V}(\vec{r}) \phi(\vec{r}) \right]$$

$$i\hbar \frac{\partial f(t)}{\partial t} = E f(t)$$

$$-\frac{1}{2} \nabla^2 \phi(\vec{r}) + \hat{V}(\vec{r}) \phi(\vec{r}) = E \phi(\vec{r}) \quad \rightarrow \quad \text{time independent Schrödinger equation}$$

in Dirac notation: $\hat{H}|\phi\rangle = E|\phi\rangle$

Postulate 5

Time independent potentials: Stationary states

$$i\hbar \frac{\partial f(t)}{\partial t} = E f(t) \quad \rightarrow \quad f(t) = e^{-iEt/\hbar} \quad \text{and hence,} \quad \psi(\vec{r}, t) = \phi(\vec{r}) e^{-iEt/\hbar}$$

Probability density does not depend on t : $|\psi(\vec{r}, t)|^2 = \phi^*(\vec{r}) e^{iEt/\hbar} \phi(\vec{r}) e^{-iEt/\hbar} = |\phi(\vec{r})|^2$

When $\hat{A}(t) = \hat{A}$, its expectation value is time-independent:

$$\langle \hat{A} \rangle_t = \langle \phi(\vec{r}) e^{-iEt/\hbar} | \hat{A} | \phi(\vec{r}) e^{-iEt/\hbar} \rangle = e^{iEt/\hbar} e^{-iEt/\hbar} \langle \phi(\vec{r}) | \hat{A} | \phi(\vec{r}) \rangle = \langle \hat{A} \rangle_{t_0}$$

Postulate 5

Time independent potentials: Non-stationary states

The general solution of the Schrödinger equation $\psi(\vec{r}, t) = \sum_i c_i \phi_i(\vec{r}) e^{-iE_i t/\hbar}$ is typically **not** stationary

linear combination of stationary functions

Probability density: $|\psi(\vec{r}, t)|^2 = \psi^*(\vec{r}, t)\psi(\vec{r}, t) = \sum_i c_i^* \phi_i^*(\vec{r}) e^{iE_i t/\hbar} \sum_j c_j \phi_j(\vec{r}) e^{-iE_j t/\hbar}$

$$= \sum_i |c_i|^2 |\phi_i(\vec{r})|^2 + 2 \sum_i \sum_{j>i} \text{Re} \left[c_i^* c_j e^{i(E_i - E_j)t/\hbar} \phi_i^* \phi_j \right]$$

For example: $\psi(\vec{r}, t) = \frac{1}{\sqrt{2}} \phi_0 e^{-iE_0 t/\hbar} + \frac{1}{\sqrt{2}} \phi_1 e^{-iE_1 t/\hbar}$

$$|\psi(\vec{r}, t)|^2 = \psi^*(\vec{r}, t)\psi(\vec{r}, t) = \left(\frac{1}{\sqrt{2}} \phi_0^* e^{iE_0 t/\hbar} + \frac{1}{\sqrt{2}} \phi_1^* e^{iE_1 t/\hbar} \right) \left(\frac{1}{\sqrt{2}} \phi_0 e^{-iE_0 t/\hbar} + \frac{1}{\sqrt{2}} \phi_1 e^{-iE_1 t/\hbar} \right)$$

$$= \frac{1}{2} |\phi_0|^2 + \frac{1}{2} |\phi_1|^2 + \frac{1}{2} \phi_0^* \phi_1 e^{i(E_0 - E_1)t/\hbar} + \frac{1}{2} \phi_0 \phi_1^* e^{-i(E_0 - E_1)t/\hbar}$$

$$z + z^* = 2 \text{Re } z$$

Postulate 5

$$e^{ix} = \cos(x) + i \sin(x)$$

$$|\psi(\vec{r}, t)|^2 = \frac{1}{2}|\phi_0|^2 + \frac{1}{2}|\phi_1|^2 + \text{Re}(\phi_0^* \phi_1 e^{i(E_0 - E_1)t/\hbar})$$

using two eigenfunctions of the harmonic oscillator: $\phi_0 = \left(\frac{\alpha}{\pi}\right)^{1/4} e^{-\alpha x^2/2}$ and $\phi_1 = \left(\frac{4\alpha^3}{\pi}\right)^{1/4} x e^{-\alpha x^2/2}$

$$|\psi(\vec{r}, t)|^2 = \frac{1}{2}\phi_0^2 + \frac{1}{2}\phi_1^2 + \phi_0\phi_1 \cos(E_0 - E_1)t/\hbar$$

